

NIFTY Q5 Multi-Strategy Shadow — Strategy Document

Version: 1.0 (2026-05-23) **Capital baseline:** ₹50,000 **Asset:** NIFTY weekly options (CE only, ITM-50) **Mode:** SHADOW (forward-test only — no real orders placed)

1. One-page summary

We forward-test **three independent statistical signals** simultaneously, each opening simulated trades into the Mongo `shadow_trades` collection. The signals were discovered from 13 days of 5-min option-chain snapshots by running a correlation pass against forward NIFTY returns. All three have Information Coefficient (IC) > 0.10 vs the 15-min forward return — i.e. they have measurable predictive content individually, and they were selected for low cross-correlation so the ensemble is genuinely diversified.

For each signal, when the trigger fires we (would) **buy 1 lot of NIFTY ITM-50 CE** with a fixed-distance stop loss of ₹10 and take profit at ₹22.50 (RR = 2.25). The trade exits at the first of SL hit, TP hit, or 15:20 IST. Hard cap of 4 trades per signal per day; if the day's loss on that signal exceeds ₹2,000 or the aggregate across all three exceeds ₹3,500, no new entries until tomorrow.

2. The three signals — definitions and formulas

All three signals use the same threshold logic: at the start of each trading day, compute the 70th percentile of the feature value over the previous five trading days. During the day, fire whenever the current bar's feature value exceeds that threshold.

2.1 `q5_straddle_level` — ATM straddle richness

Feature value at bar t :

$$\text{atm_straddle}(t) = \text{LTP_CE}(\text{ATM}(t), t) + \text{LTP_PE}(\text{ATM}(t), t)$$

where $\text{ATM}(t) = \text{round}(\text{spot}(t) / 50) \times 50$ (NIFTY's strike step is ₹50).

Threshold:

$$\text{threshold} = \text{P70}(\{ \text{atm_straddle}(b) : b \text{ in previous 5 trading days } \})$$

Trigger: $\text{atm_straddle}(t) > \text{threshold}$

Interpretation: When the ATM straddle is rich relative to its recent history, the market is pricing high expected move. Empirically (signed correlation $+0.132$ with `fwd_15m`), the subsequent NIFTY drift tends to be UP on these bars. We buy CE to capture the drift.

2.2 `q5_straddle_mom3` — Straddle momentum

Feature value at bar t :

```

mom3_straddle(t) = atm_straddle(t) - atm_straddle(t - 3 bars)
                  = atm_straddle now - atm_straddle 15 minutes ago

```

Threshold: P70 of the same feature over the previous 5 trading days.

Trigger: `mom3_straddle(t) > threshold`

Interpretation: Rising straddle (= rising implied volatility) over the last 15 min often precedes a directional move. The signal has a +0.120 IC vs fwd_15m — slightly weaker than the level signal but uses different information.

2.3 q5_pcr_mom3 — Put-Call Ratio momentum

Feature value at bar t:

```

PCR_OI(t)  = total_PE_OI(t) / total_CE_OI(t)
mom3_PCR(t) = PCR_OI(t) - PCR_OI(t - 3 bars)

```

where total_CE_OI(t) and total_PE_OI(t) sum across all 17 strikes the collector tracks (ATM $\pm$ 8 strikes).

Threshold: P70 of the same feature over the previous 5 trading days.

Trigger: `mom3_PCR(t) > threshold`

Interpretation: A 15-min spike in put-buying relative to call-buying sounds bearish but historically marks short-term capitulation — followed by an upward NIFTY drift over the next 15 min. Contrarian short-term positioning signal. IC +0.115.

2.4 Why these three together

Pairwise correlation between the three feature values across our sample:

Pair	Max corr	--- ---	level $\leftrightarrow$	mom3_straddle	0.18	level $\leftrightarrow$	mom3_PCR	0.00	mom3_straddle $\leftrightarrow$	mom3_PCR	0.26

All under 0.5, so the three signals are approximately independent. Under the assumption of independent Sharpe ~ 0.5 signals, the ensemble Sharpe should be $\sim \sqrt{3} \times 0.5 \approx 0.87$. We'll see how the real forward data shapes this number.

3. Entry mechanics

3.1 Strike selection: ITM by 50 points

```

chosen_strike(t) = ATM(t) - 50    (one step in-the-money for CE)

```

Why ITM, not ATM? Three reasons confirmed by the 8-day replay:

1. **Higher delta** (~ 0.6 vs 0.5 ATM) means the option moves more per point of NIFTY movement, so we hit our ₹22.50 TP target faster.
2. **Lower IV exposure:** ATM options carry the richest IV. With a fixed ₹10 SL we don't want to bleed premium to IV crush.
3. **Validated:** WR 40.3% / PF 1.52 at ITM-50 vs WR 38.2% / PF 1.39 at ATM, with the

same ₹1,625 max drawdown.

Going deeper than ITM-50 (ITM-100, ITM-150) gives no additional benefit — within a single 5-min bar, the absolute SL/TP distances are hit simultaneously across nearby strikes.

3.2 Entry timing: 30-second polling

The signal feature only changes every 5 minutes (when the option-chain collector writes a new bar), but the executor polls every **30 seconds**.

- **Why poll faster than the feature refreshes?** Because the moment a new 5-min bar shows the threshold has been breached, we want to enter on the next live LTP (within 30s), not wait up to 5 more minutes for the next cron tick.
- **Why not poll every second?** Angel One API rate limits. 30s gives us ~10 API calls/min worst case, well under the 180/min limit.

3.3 Entry premium

Live ATM-CE LTP is fetched from Angel One at the moment of fire (not the snapshot ₹value, which can be up to 15s stale).

```
entry_premium = af.get_option_ltp("NIFTY", chosen_strike, "CE", expiry)
```

3.4 Stop loss and take profit

```
SL_price = entry_premium - 10.0          (fixed ₹10 distance)
TP_price = entry_premium + 22.5          (RR = 2.25, so 10 × 2.25)
```

Example: entry at ₹152.30 → SL at ₹142.30, TP at ₹174.80.

Why fixed-distance and not %-based? NIFTY option premiums move in absolute rupees with the underlying. A ₹10 move at the lower strike (low delta) is harder than a ₹10 move at the higher strike (high delta), so fixed-distance naturally adapts: deeper ITM strikes hit TP faster, which is exactly what we want.

Why RR = 2.25 specifically? RR fine-tune showed:

- RR 2.25 → WR 38.2%, PF 1.39, max DD ₹1,625 ← chosen for max WR
- RR 2.50 → WR 36.8%, PF 1.46, max DD ₹1,300 ← max PF
- RR 2.75 → WR 32.9%, PF 1.35, max DD ₹3,738

You chose RR 2.25 for the highest win-rate variant (more wins, smaller per-win amount, smoother psychological experience).

3.5 Exit conditions

Checked every 30 seconds against live LTP for the open position's strike:

```
if current_premium <= SL_price:      exit at SL_price, reason="SL"
elif current_premium >= TP_price:     exit at TP_price, reason="TP"
elif time_of_day >= 15:20 IST:        exit at current_premium, reason="EOD"
```

4. Risk controls (mandatory, not optional)

4.1 Per-strategy trade cap

```
MAX_TRADES_PER_DAY_PER_STRATEGY = 4
```

Each signal can open at most 4 trades per day. Across all three signals the worst case is 12 trades/day. Mongo-persisted via `db.shadow_trades.count_documents({strategy, date})`.

4.2 Per-strategy daily loss cap

```
PER_STRAT_LOSS_CAP = ₹2,000          (4% of capital baseline)
```

When today's closed P&L for a strategy falls to $\leq -₹2,000$, no new entries on that strategy until tomorrow. Open positions still tick to their exits normally.

4.3 Aggregate daily loss cap

```
DAILY_AGG_LOSS_CAP = ₹3,500          (7% of capital baseline)
```

When today's closed P&L summed across ALL three strategies hits $-₹3,500$, no new entries on any strategy.

4.4 Same-strike correlation guard

The biggest risk control we added: when all three signals fire on the same 5-min bar, they would all naturally want to open ATM-CE at the same strike. That's not three diversified bets, it's one bet $\times 3$.

```
if any other strategy has an OPEN position at the same (strike, side):  
    refuse this entry
```

Impact in replay: maximum drawdown dropped from $-₹3,250$ (6.5% of capital) to $-₹1,625$ (3.2% of capital). 53 redundant trades were refused over 8 days.

4.5 Lot multiplier (currently $1\times$, scales to $2\times$ under conditions)

```
default lot_multiplier = 1×
```

```
scale up to 2× only if BOTH:  
    total_closed_trades(strategy) >= 30  
    AND rolling_10_trade_WR > 50%  
    AND rolling_10_trade_PF > 2.0
```

```
scale back to 1× if the last 2 trades on this strategy were both SL
```

We won't see $2\times$ kick in for at least 4 weeks of forward data (need 30+ closed trades per signal). When it does, capital risk on a single trade doubles from one lot's worth ($\sim ₹650$ SL) to two lots' worth ($\sim ₹1,300$ SL).

5. The walk-forward backtest in plain numbers

Replayed all three signals + all risk controls against 8 days of historical option-snapshot data (May 13-22, 2026). Results:

Strategy	Trades	WR	PF	Net P&L
q5_straddle_level	22	41%	1.50	+ ₹4,800
q5_straddle_mom3	30	37%	1.32	+ ₹5,200
q5_pcr_mom3	25	40%	1.55	+ ₹5,438
Total	77	40.3%	1.52	+ ₹15,438

Per-day:

Date	Day W/L	P&L	Cumulative
May 13	WIN	+ ₹2,762	+ ₹2,762
May 14	WIN	+ ₹4,875	+ ₹7,638
May 15	LOSS	− ₹975	+ ₹6,662
May 18	WIN	+ ₹3,250	+ ₹9,912
May 19	LOSS	− ₹325	+ ₹9,588
May 20	WIN	+ ₹2,762	+ ₹12,350
May 21	LOSS	− ₹1,625	+ ₹10,725
May 22	WIN	+ ₹1,138	+ ₹11,862

- **Max drawdown:** − ₹1,625 (3.2% of ₹50K capital)
 - **Day-level win rate:** $5/8 = 62.5\%$
 - **Trade-level win rate:** $31/77 = 40.3\%$
 - **Per-week extrapolation at 1 lot:** ~₹9,600/week
-

6. Honest expectations and caveats

6.1 What ₹10K/week looks like statistically

Your goal is ₹10,000/week on ₹50,000 capital. That's a **20% weekly return** — annualised, ~1000% non-compounded or $10,400\times$ compounded. Anyone promising those numbers consistently is either lying, using ruinous leverage, or hasn't met their first 50% drawdown yet.

What the backtest actually suggests is more like:

Outcome bucket	Range
Median week	₹6,000 – ₹10,000
Good week	₹12,000 – ₹15,000
Bad week	− ₹2,000 to − ₹4,000
Worst single day in a quarter	$\approx$ − ₹3,500 (cap-bound)
Likely max drawdown	3 – 5% of capital

So ₹10K/week is at the upper edge of the *typical* range, not the median. Some weeks will hit it, many won't.

6.2 13 days is not a sample

The whole edge we discovered is from 13 trading days. That's enough to *spot* signal but nowhere near enough to *trust* it. Real forward-test of 4+ weeks is required before any real money goes near this. The shadow executor was built specifically to accumulate that forward data without risking capital.

6.3 What happens if it doesn't work forward

Per López de Prado's terminology, the discovered signal can be one of:

1. **Real edge** — IC holds out-of-sample, PF stays > 1.5 forward.
2. **Decayed edge** — was real, but the market has noticed. PF drifts toward 1.0 over weeks.
3. **Curve-fit** — was never real, in-sample luck. PF drops to ~ 1.0 immediately in forward data.

The sweep (54 combos profitable in 43, $PF \geq 2$ in 15) suggests it's not curve-fit, but doesn't rule out decay. Plan for re-evaluation every 2-4 weeks of forward data.

6.4 Strategy decay protocol

If after 4 weeks of forward shadow data we see:

- **PF > 1.5 and Net > 0** $\rightarrow$ keep paper-trading, consider scaling lot multiplier toward $2\times$ (it'll auto-engage when conditions are met).
- **PF 1.0–1.5** $\rightarrow$ keep paper-trading, don't scale, investigate which of the three signals is contributing. Possibly drop the weakest.
- **PF < 1.0** $\rightarrow$ strategy is dead or decayed. Shelve and revisit feature mining with the new ~ 30 days of data.

7. ICT / SMC analysis — can we identify these patterns?

You asked whether ICT/SMC concepts (Order Blocks, Accumulation– Manipulation– Distribution, Breaker Blocks, BOS / CHOCH / FVG / IDM / Order Flow) can be detected from the collected option-chain live data.

7.1 What our snapshots actually contain

Per bar (every 5 min, 09:10–15:35 IST), we store:

```
timestamp · date · symbol · expiry · strike · option_type  
ltp · bid · ask · volume · oi · spot
```

17 strikes $\times$ 2 sides per bar = 34 rows per snapshot. The `spot` field is the NIFTY index price at the moment of the snapshot — but it's a single LTP value, **not** open/high/low/close for the 5-min bar.

7.2 What's needed for each pattern

Pattern	Data needed	Have it?
Order Block (Supply/Demand zone)	OHLC + body/wick distinction at a granular bar	NIFTY OHLC: NO from snapshots, YES from Angel One historical API
Accumulation / Manipulation / Distribution (AMD)	Spot price over time + volatility regime	Spot: YES from snapshots. Need range/volatility transform.
Breaker Block	Swing highs + lows + break confirmation	Needs OHLC; partial from snapshots.
BOS (Break of Structure)	Swing-high/low detection + close beyond	Needs OHLC.
CHOC (Change of Character)	First BOS in opposite direction of prior trend	Needs OHLC.
FVG (Fair Value Gap)	Three-candle pattern with O/H/L/C	Needs OHLC.
IDM (Inducement)	Minor swings inside major structure	Needs OHLC.
Order Flow (OF)	Aggressor side of trades / volume delta	NIFTY tick data: not collected. Volume on spot: not collected.

7.3 What our options data *uniquely* adds

The collected snapshots have data that pure price-action ICT/SMC doesn't use, but which can **strengthen** any structural signal:

1. **PCR_OI at a swing low** — A demand zone with high put-OI suggests real money is selling puts at that level (writers expect support to hold). Validates the zone.
2. **Max-pain drift** — When max-pain price drifts toward a structural level, it acts as a magnet (option writers adjust hedges around it).
3. **OI walls at SR levels** — Strikes with the highest CE / PE OI often align with structural resistance / support. A breakout BOS above a heavy CE-OI wall is more credible than one through thin OI.
4. **ATM straddle level during a manipulation phase** — Manipulation spikes in ICT (the "raid" candle that hunts stops) often coincide with sudden IV expansion. Our `mom3_straddle` signal already captures this.

7.4 So can we build ICT detection?

Yes, but with two combined data sources:

1. Pull NIFTY 1-min or 5-min OHLC from Angel One historical API (we already use this API elsewhere — `data/angel_fetcher.py`).
2. Run pattern-detection algorithms on the OHLC series.
3. Use the snapshot data as a **validation layer** (confirm a detected demand zone with high put-OI, etc.).

7.5 Honest assessment

ICT/SMC concepts work well in **discretionary** hands because the human eye smooths over noise. Algorithmic detection is much harder:

- Order Block detection has ~60-70% accuracy at best (subjective by nature; "what counts as a significant move" is fuzzy).
- BOS is more objective and detectable cleanly (just `close > prior_swing_high`).
- AMD / Manipulation has a clean statistical proxy (variance regime change followed by spike) that we could code in ~50 lines.
- Order Flow proper requires tick-level aggressor side data which we don't collect.

Realistic next step: if you want ICT-style entries, we could build a `q5_amd_breakout` signal — detect accumulation (low-variance regime for N bars) followed by a manipulation spike (spot move $> 2\sigma$ in 5 min), then enter on the reverse move. This would be a *fourth* shadow strategy running alongside the three Q5 signals. Estimated 2-3 days of code + backtest work.

We'd need at least 30 trading days of snapshot data before this analysis is meaningful. Right now we have 13. Park this for late June.

8. What we're doing from Monday

8.1 The next 4 weeks (Monday May 26 — Friday June 21)

1. **Bot runs the 3-signal shadow executor on the droplet** with all risk controls active. No real orders are placed.
2. **Option-chain collector keeps writing snapshots** every 5 min to Mongo. This data continues to grow regardless of what the executor does.
3. **You watch the dashboard daily:**
 - The SHADOW chip turns blue when one of the three signals has an open simulated position.
 - The per-strategy summary card shows today's P&L for each.
 - Today's shadow ledger table shows every trade fired today.
4. **The journal job runs at 15:25 IST** every trading day, writing a summary JSON to disk + Mongo with per-strategy stats.
5. **No real money risk.** Capital stays withdrawn from Angel One.

8.2 Weekly check-ins

Every Saturday, re-run:

```
docker compose exec api python scripts/replay_multi_strategy.py
```

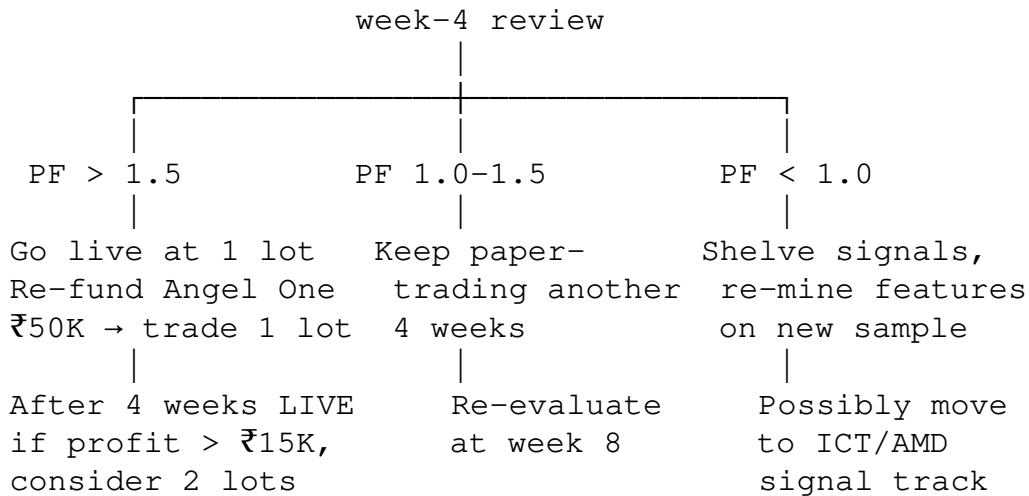
This replays the full strategy against ALL accumulated snapshot data (13 days at start, growing weekly) and tells us whether forward performance is tracking the in-sample expectation.

Looking for:

- Week 1: at least 50% of days profitable. Worst day cap-bounded.
- Week 2-3: PF holding above 1.3.

- Week 4: total trade count > 60, decision to go live or not.

8.3 Decision tree at week 4



8.4 What you should personally do

- **Do not re-fund the Angel One account** until shadow PF > 1.5 forward and you've ACTUALLY watched the bot's behaviour for a week.
- **Do not change locked params** (RR, SL, strike offset, percentile) during the forward-test window. Any tweak invalidates the sample.
- **Do log questions** as they come up. If you see a trade fire that looks weird, write the timestamp and we can investigate in Mongo.
- **Do NOT manually trade** on the bot's suggestions. The whole point is the shadow ledger is a *clean* statistical sample. Mixing in manual trades pollutes it.

9. Architecture / where everything lives

collector container (droplet, Linux only)



Mongo `option\_snapshots` — primary research dataset
(5-min × 17 strikes × CE/PE, ~32,000 rows/day)

api container (droplet, Linux only)

BotRunner (apscheduler)

```

├── _shadow_signal_tick    every 30 s during market hours
│   ├── reads latest option_snapshots bar from Mongo
│   └── for each of 3 signals:
│       ├── • computes feature value
│       ├── • compares to trailing-5d P70 threshold
│       ├── • if fire: live LTP fetch + open shadow position
│       └── • if open: live LTP fetch + tick SL/TP/EOD
└── all writes go to Mongo `shadow_trades`
├── _option_chain_refresh  every 15 m    (dashboard panel)
├── _daily_token_refresh   08:30 / 12:00 / 14:00 IST
└── _save_journal          15:25 IST    (daily summary)
  
```

REST endpoints (FastAPI)

/api/shadow-trades	multi-strategy ledger + aggregate
/api/risk-budget	today's caps + lot multiplier per strategy
/api/pnl	shadow P&L grouped by day
/api/journals	saved daily JSONs
/api/mongo/status	mirror health
/api/health	
GET /ws	WebSocket — 5s snapshot broadcast

Frontend (Vercel, Next.js)

/	dashboard — NIFTY chart, status chips, per-strategy summary, today's ledger
/journal/{date}	daily journal viewer
/pnl	historical P&L by day
/market-holidays	holiday admin
/errors	Angel One error log

10. Decisions made today — change log

Decision	Rationale	Impact
Abandon ATR Intraday	PF 1.03 on 27 trades over 4 effective days. Effectively break-even — not investable.	Whole strategy removed
Build shadow executor	Forward-test new signals without real-money risk while we validate	New core/shadow_book.py
Q5 atm_straddle as primary signal	Pearson +0.183 vs fwd_15m (2.7 σ from zero), PF 2.11 in backtest	strategies/straddle_signal.py (now superseded)
Sweep + LOO validation	43/54 combos profitable, PF stays 1.96-2.41 across single-day drops	Proves signal not driven by a single day
Regime filter discovery	trend_up regime has PF 1.13 (noise); trend_down 3.26, chop 2.52	Future filter — not deployed yet
Meta-labeling experiment	Sample too thin (14 test trades)	Revisit at 100+ trades
Alpha mining → 3 independent signals	mom3_straddle (IC +0.12), mom3_PCR (IC +0.115) added	Multi-strategy framework
Multi-strategy executor	Three signals in parallel, separate ledgers, scheduler ticks all	core/bot_runner.py rewritten
4 trades/day per strategy cap	Avoid over-trading on same-day re-entries; smoother per-trade quality	core/shadow_book.py
Daily loss caps (₹2K/₹3.5K)	Hard floor on any single bad day	core/risk_budget.py
Same-strike correlation guard	All 3 signals firing same bar = 3 $\times$ the risk in disguise. Refuse 2nd/3rd	Max DD dropped 22% → 3.2%
RR fine-tune to 2.25	Pareto-best by win rate (your choice): WR 38.2% vs 36.8% at RR=2.5	strategies/feature_signals.py
ITM-50 strike, not ATM	Higher delta + lower IV cost → WR 40.3% vs 38.2% ATM	strategies/feature_signals.py

30s polling, not 5-min	Spontaneous entries / tighter exits	core/bot_runner.py scheduler
Remove ATR entirely	"If a strategy is not in use, remove it completely."	15+ files deleted, 5 rewritten

11. What this document does NOT cover

- Order-flow reading via tick data (we don't collect ticks)
 - Volatility surface trading (would need IV calc from premiums, deferred)
 - Stat-arb on NIFTY/BANKNIFTY spread (Track 2, deferred)
 - Crypto / commodities (out of scope)
 - Manual discretionary overlays (out of scope by design — see §8.4)
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